

Assignment 5 Tables

Network Interface Information

Node	Interface	IP Address	MAC Address
Thor	eth1	10.10.1.100/24	02:56:a2:90:1a:a0
Router	eth1	10.10.1.1/24	02:fa:5c:26:a1:f5
Router	eth2	10.10.2.1/24	02:4f:46:a8:a9:ec
Loki	eth1	10.10.2.100/24	02:0c:81:a6:86:b6

Network Topology

Source			Destination		
Node	Interface	IP	Node	Interface	IP
Thor	eth1	10.10.1.100/24	Router	eth1	10.10.1.1/24
Router	eth1	10.10.1.1/24	Thor	eth1	10.10.1.100/24
Router	eth2	10.10.2.1/24	Loki	eth1	10.10.2.100/24
Loki	eth1	10.10.2.100/24	Router	eth2	10.10.2.1/24

Network Throughput

Throughput Table

Time Interval	% Packet Loss
1.2	83.3%
1.1	73.3%
1.0	
0.825	

0.800	
0.775	
0.750	
0.725	
0.700	
0.675	
0.650	
0.600	
0.500	
0.400	

Throughput Graph

Time Interval (seconds) and % Packet Loss



Analysis

- a. Estimated Maximum Throughput: $30 \text{ packets} / 0.8 \text{ seconds} = 37.5 \text{ packets/sec}$
- b. I did observe rampant packet loss. The most likely explanation for this packet loss is network congestion - since the devices in this LAN were provisioned on a cloud-based system, poor allocation of resources to the router might have caused it to be unable to handle a high volume of traffic from between Thor and Loki. Therefore, any overflowing packets would have to be dropped. This also explains the absurdly high level of packet loss across all time intervals - the router was most likely not provisioned with enough resources to handle heavy incoming and outgoing traffic.